

# Runye (Ray) XU

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## EDUCATION

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**The Chinese University of Hong Kong (CUHK)** *Sep. 2024 - Nov. 2025*

*Master of Science in Financial Technology*

- **Main Courses:** Quantitative Risk Management, Quantitative and Algorithmic Trading, Data Analytics for Financial Technology, Applied Blockchain and Cryptocurrencies

**The Chinese University of Hong Kong, Shenzhen (CUHKSZ)** *Sep. 2020 - Jul. 2024*

*Bachelor of Business Administration in Financial Engineering*

- **Main Courses:** Asset Pricing, Fixed Income Securities Analysis, Options and Futures, Financial Data Analysis, Financial Computation, Mathematical Modeling, Optimization, Regression Analysis, Stochastic Processes, Statistical Computing, Data Structures, Machine Learning

**Sophia University (上智大学)** *Mar. 2023 - Aug. 2023*

*School-Nominated Exchange Program at Tokyo, Japan*

## SCHOLARSHIPS & AWARDS

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**CUHKSZ Bowen Scholarship (University Level)** *2020 – 2024*

RMB 30,000 per academic year

## RESEARCH EXPERIENCE

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**The Chinese University of Hong Kong | Research Assistant (FT)**

*Prof. Paul Whelan* *Sep. 2025 - Present*

- **Faculty:** Department of Finance **Field:** Empirical Asset Pricing
- **Main Jobs:** literature review; data fetching, cleaning and analyzing using Python & MATLAB; report writing

**The Chinese University of Hong Kong, Shenzhen | Research Assistant (PT)**

*Prof. Bohui Zhang* *Sep. 2022 - Nov. 2022*

- **Faculty:** School of Management and Economics **Field:** Financial Technology
- **Main Jobs:** empirical cases research; data collection and cleaning using Python; presentation slides preparation

## RESEARCH PROJECTS

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**The Price of Duration Risk in Emerging Markets** *Ongoing*

*Personal Project, Supervisor: Prof. Paul Whelan*

- Building a security-level panel of emerging-market sovereign bonds from LSEG, spanning over 1995~2025.
- Constructing maturity-weighted debt-to-GDP (MWD/GDP) and other metrics to measure duration supply.
- Extending Greenwood and Vayanos' (2014) duration-supply framework and models to the Emerging Market.
- Examining how inflation, exchange-rate exposure will affect the original pricing and return models.

**Cryptocurrency Overnight Drift** *Ongoing*

*Coauthor: Prof. Paul Whelan, Dr. Niklas Kroner, Ms. Linda Lee Zhang*

- Developed a reproducible Python pipeline for five-minute cryptocurrency spot data from Binance, Coinbase, and OKX, covering multiple assets and quote currencies.
- Analyzed intraday return seasonality, hourly decompositions, and Newey–West-adjusted inference; identified a robust concentration of positive BTC and ETH returns around 5:00–6:00 p.m. ET.
- Conducted cross-exchange robustness tests and generated result figures, LaTeX tables, and validation reports.

**Crypto Spot & Perpetual Futures Arbitrage on Binance.** *Oct. 2024 - Apr. 2025*

*CUHK Industrial Project, Supervisor: Prof. Paul Whelan & Mr. Jason NG*

- Retrieved and processed hourly OHLCV data (2020–2025) via Binance APIs; constructed spread time series.
- Implemented a mean-reversion signal engine using rolling statistics (mean/std window) for alpha generation.
- Built a Python-based back testing framework with position management and capital constraints.
- Simulated trading performance and extended to include funding rate modelling in cash-and-carry strategy.

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## Bitcoin Volume Prediction using LSTM with Attention

Jan. 2025 - Apr. 2025

### CUHK Course Project

- Developed and hyperparameter tuned a LSTM model with attention for intraday BTC volume forecasting.
- Engineered 150+ time-series features including technical indicators (SMA, EMA, MACD, RSI) and volume-based signals (VPT, OBV, price impact); and applied PCA for dimensionality reduction.
- Designed a custom loss function (MSE + L1 + log error) to stabilize training under extreme volatility.

## INDUSTRIAL EXPERIENCE

### SDIC Securities (Hong Kong) Limited

Jan. 2026 - May. 2026

#### Fixed Income Sales and Trading Intern (Part-time)

Hong Kong SAR

- Cleaned and maintained trading records and position data in accordance with accounting standards.
- Built an automated position monitoring system to identify bonds with anomalous spreads or price movements.
- Developed a tracking system for credit spreads of IG USD financial bonds and prepared weekly credit reports.
- Produced analytical reports regarding various fixed income topics, including convertible bonds and MBS.

### HSBC

May. 2025 - Aug. 2025

#### Emerging Talent Intern

Guangzhou, China

- Participated in project management for the DSP 3.0 system within various teams in the technology department.
- Supported Non-Functional Requirement (NFR) management and streamlined project procedures.
- Conducted budget tracking, variance analysis and re-planning across multiple technical lines under the project.

### Changjiang Securities

May. 2024 - Aug. 2024

#### Industry Research Intern

Shanghai, China

- Updated and maintained database of Chinese Electric Utility Industry and did data visualization in Excel.
- Contributed to daily, monthly reports and in-depth analysis of Electric Utility Industry and listed companies.

### BridgeOne Capital

Sep. 2023 - Dec. 2023

#### Investment Assistant Intern

Shenzhen, China

- Tracked and analyzed shipment and price data of lithium batteries, photovoltaic modules, and related materials.
- Prepared meeting minutes and helped complete Market Insight Report regarding the New Energy Industry.
- Conducted research and fundamental analysis on startups in hard-tech sectors, including robotics and aerospace.

## SKILLS & CERTIFICATES

- **Programming:** Python, MATLAB, R, STATA, MySQL, q/kdb+, Java
- **Research Tools:** LaTeX, Bloomberg, Refinitiv Workshop (GUI/API), CSMAR, iFinD
- **Language:** Mandarin (Native), English (GRE 328; IELTS 7.0), Japanese (JLPT N2), Cantonese (Basic)
- **Qualification:** FRM Part I